

Maths for IT - II:-

Queuing Theory:-

λ (lambda): Avg arrival rate.

μ : Avg service Rate.

ρ : Traffic Intensity / Utilization factor.
(λ/μ)

$$\rho = \frac{\lambda}{\mu}$$

Operating Characteristics for M/M/1 Model:- (Infinite Capacity)

- Avg No. of Customers in system (L) = $\frac{\lambda}{\mu - \lambda}$.
- Avg No. of customers in queue (L_q) = $\frac{\lambda^2}{\mu(\mu - \lambda)}$.
- Avg time a customer spends in system (W) = $\frac{1}{\mu - \lambda}$.
- Avg. Waiting time in Queue = $\frac{\lambda}{\mu(\mu - \lambda)}$.
(W_q)
- Probability of Zero customers in system (P_0) = $1 - \frac{\lambda}{\mu} = 1 - \rho$.
- Probability of 'n' customers in system = $(1 - \rho) \times \rho^n$.
(P_n)
- Probability that waiting time in system exceeds t = $e^{-(\mu - \lambda)t}$.
($P(W > t)$)

Operating Characteristic for M/M/1 : N Model (finite Capacity)

- Probability of zero customers (P_0) = $\frac{1 - \rho}{1 - \rho^{N+1}}$.

$$\text{Probability of 'n' customers in system } (P_n) = \frac{1 - \rho}{1 - \rho^{N+1}} \times \rho^n$$

- Probability that system is full $= P_0 \times e^N$
(P_N)

- Effective Arrival Rate (λ_e) $= \lambda \times (1 - P_N)$

- Average number of customers in system (L)

$$= \frac{e \times (1 - (N+1)e^N + Ne^{N+1})}{(1-e)(1-e^{N+1})}$$

- Average Waiting time in system (W) $= \frac{L}{\lambda_e}$

* Relationship b/w Measures :-

(Little's formulae)

Average Number in Queue (L_q) $= \lambda \times W_q$

Average Waiting time in Queue (W_q) $= \frac{L_q}{\lambda}$

INVENTORY CONTROL:-

Basics

- Ordering Cost per unit: $\frac{C_3}{q}$
- Average Inventory: $\frac{\text{Beginning Inventory} + \text{Ending Inventory}}{2}$

- Total Carrying Cost: $\frac{q}{2} \times C_1$

Carrying cost as a % $\rightarrow \frac{q}{2} \times i \times P$

Carrying cost / unit = $i \times P$
(C₁)

$q \equiv$ Order Quantity
(or lot size)

$P \equiv$ Purchase price
(production cost / time)

- Number of orders: $\frac{\lambda}{P} \left(\frac{D}{q_0} \text{ or } \frac{x}{q} \right)$

D, r, λ : Demand rate or annual demand.

- Total Ordering Cost: $\frac{D}{q} \times C_3 \text{ or } \frac{\lambda}{q} \times C_3$

- Total Inventory cost (excluding material cost): $C_q = \frac{1}{2} q C_1 + \frac{C_3 x}{q}$

- Total Cost: Ordering Cost + Carrying Cost + Material cost.
(including Material cost)

$$= \left(\frac{\lambda}{q} \times C_3 \right) + \left(\frac{q}{2} \times i \times P \right) + P \lambda$$

- Annual Consumption Value: $q \times P$
(for ABC Analysis)

* Economic Order Quantity (EOQ) / Economic Lot Size Model

- Basic EOQ / Economic Batch Quantity (EBQ): $q_0 = \sqrt{\frac{2C_3 r}{c_1}}$

- EOQ (Purchase Model with Carrying Percentage): $q_0 = \sqrt{\frac{2C_3 r}{i \times p}}$

- Optimal Inventory Cost $(C_0) = \sqrt{2C_3 r}$
 $= \sqrt{2C_3 \times i \times p \times \lambda}$

- Total Optimal Cost: $\sqrt{2C_3 \times i \times p \times \lambda} + p \lambda$
 (including Material)

- Optimal Time b/w orders $(t_0) = \frac{q_0}{r}$

$t_0 = \sqrt{\frac{2C_3}{c_1 r}}$ or $\sqrt{\frac{2C_3}{i \times p \times \lambda}}$

- Optimum No. of orders $(N_0) = \frac{\lambda}{q_0}$

- No. of Days per order $(d) = \frac{365}{N_0}$

* EOQ Model with Shortages Allowed: -

- Optimal Order Quantity $(q_0) = \sqrt{\frac{2C_3 D}{c_1} \times \frac{c_1 + c_2}{c_2}}$

- Max^m Shortage Level $(s) = \left(\frac{c_1}{c_1 + c_2} \right) \times q_0$

- Max^m Inventory Level: $q_0 - s$

Programme Evaluation & Review Technique:-

Critical Path Method:-

Pert Time Estimates & Probability Formulas:-

- Expected Time (t_e) = $\frac{t_o + 4t_l + t_p}{6}$

t_o $\equiv$ Optimistic Time. t_p $\equiv$ Pessimistic Time.
 t_l $\equiv$ Most Likely Time.

- Standard Deviation of an Activity (σ) = $\frac{t_p - t_o}{6}$

- Variance of an Activity (σ^2) = $\left(\frac{t_p - t_o}{6}\right)^2$

- S.D of project (σ) = $\sqrt{\sum \sigma^2}$

Sum of variances of all critical activities.

- Z-value (Calculate Completion Probability)

$$Z = \frac{T_z - T_e}{\sigma}$$

T_z $\equiv$ Contractual (target) Project completion Time.

T_e $\equiv$ Expected project completion Time.

* Network Analysis & Float (Slack) Formulas:-

- Slack (τ) = $T_L - T_E$

- Total float (τ_F) = Latent Start Time - Earliest Start Time.
(LS) (ES)
= Latest finish Time - Earliest Finish Time
(LF) (EF)

- Free float (τ_{FF}) = Earliest start of Next Activity - Earliest finish of Current Activity.

- Independent float (τ_F) = free float - slack of tail event.

T_E : Earliest Event Time.

T_L : Latest Event Time.

* Crashing (Time - Cost Trade-off)

$$\text{Cost slope} = \frac{\text{Crash Cost} - \text{Normal Cost}}{\text{Normal Time} - \text{Crash Time}}$$

$$\text{Total Project Cost} = \text{Direct Cost} + \text{Indirect Cost}$$

$$\frac{0.1 - 0.5}{-1} = (-0.4)$$

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the following are
critical activities

$$\frac{0.1 - 0.5}{-1} = (-0.4)$$

value (value)
(probability)

and the project (target) value

and the project (target) value